Experiment 6

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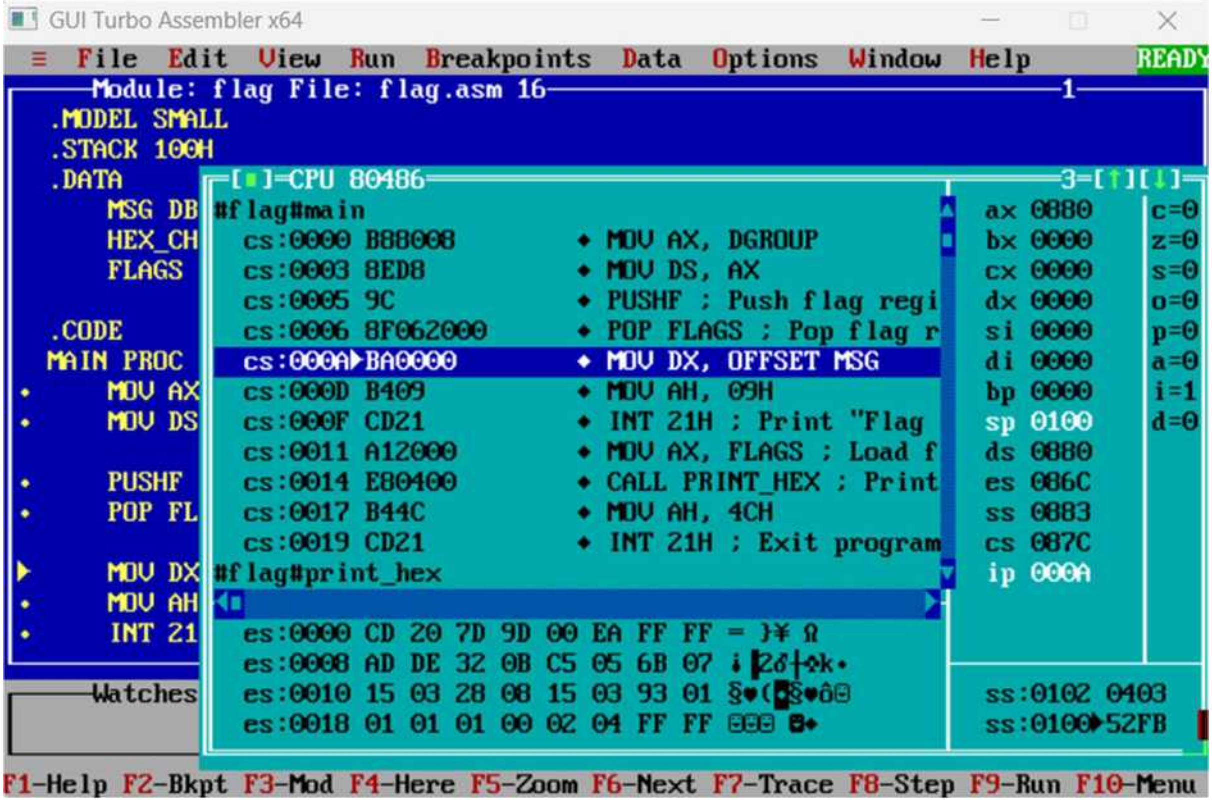
```
.MODEL SMALL
.STACK 100H
.DATA
    MSG DB 'Flag Register: $'   ; Message to display
    HEX_CHARS DB '0123456789ABCDEF'  ; Lookup table for hex digits
    FLAGS DW ?   ; Variable to store flag register value

.CODE
MAIN PROC
    MOV AX, DGROUP
    MOV DS, AX

    PUSHF         ; Push flag register onto the stack
    POP FLAGS      ; Pop flag register into FLAGS variable

    MOV DX, OFFSET MSG
    MOV AH, 09H
    INT 21H       ; Print "Flag Register: "

    MOV AX, FLAGS  ; Load flag register value into AX
    CALL PRINT_HEX ; Print the flag register in hexadecimal format

    MOV AH, 4CH
```

```
    INT 21H       ; Exit program
MAIN ENDP

; ---------------------------
; Print 16-bit Hex Procedure
; ---------------------------
PRINT_HEX PROC
    MOV CX, 4      ; We have 4 hex digits (16-bit / 4-bit each)
    MOV BX, 12     ; Bit shift amount (12, 8, 4, 0)

HEX_LOOP:
    MOV DX, AX     ; Copy AX value
    MOV CL, BL     ; Move shift count into CL (Fix for SHR error)
    SHR DX, CL     ; Shift right to isolate one hex digit
    AND DX, 0FH    ; Mask the lower 4 bits
    MOV SI, DX     ; Move index to SI
    MOV DL, [HEX_CHARS + SI] ; Convert to ASCII hex character
    MOV AH, 02H
    INT 21H       ; Print the hex digit
    SUB BX, 4      ; Move to the next hex digit
    LOOP HEX_LOOP  ; Repeat until all digits are printed

    RET
PRINT_HEX ENDP

END MAIN
```

OUTPUT:


GUI Turbo Assembler x64
File Edit View Run Breakpoints Data Options Window Help READY
Module: flag File: flag.asm 16
.MODEL SMALL
.STACK 100H
.DATA
MSG DB
HEX_CH
FLAGS
.CODE
MAIN PROC
MOV AX
MOV DS
PUSHF
POP FL
MOV DX
MOV AH
INT 21
Watches
CPU 80486
#flag#main
cs:0000 B88008 MOV AX, DGROUP
cs:0003 8ED8 MOV DS, AX
cs:0005 9C PUSHF ; Push flag regi
cs:0006 8F062000 POP FLAGS ; Pop flag r
cs:000A BA0000 MOV DX, OFFSET MSG
cs:000D B409 MOV AH, 09H
cs:000F CD21 INT 21H ; Print "Flag
cs:0011 A12000 MOV AX, FLAGS ; Load f
cs:0014 E80400 CALL PRINT_HEX ; Print
cs:0017 B44C MOV AH, 4CH
cs:0019 CD21 INT 21H ; Exit program
#flag#print_hex
es:0000 CD 20 7D 9D 00 EA FF FF
es:0008 AD DE 32 0B C5 05 6B 07
es:0010 15 03 28 08 15 03 93 01
es:0018 01 01 01 00 02 04 FF FF
ax 0880 c=0
bx 0000 z=0
cx 0000 s=0
dx 0000 o=0
si 0000 p=0
di 0000 a=0
bp 0000 i=1
sp 0100 d=0
ds 0880
es 086C
ss 0883
cs 087C
ip 000A
ss:0102 0403
ss:0100 52FB
F1-Help F2-Bkpt F3-Mod F4-Here F5-Zoom F6-Next F7-Trace F8-Step F9-Run F10-Menu